

AI Glasses Carrier V1 - Parts Specification

AI Glasses Carrier V1 - Parts Specification

Date

2026-06-30

- 1. Compute module + wireless (Wi-Fi / Bluetooth)**
- 2. Camera**
- 3. Audio (amplifier, microphones, speaker)**
- 4. GNSS (positioning)**
- 5. Sensors (IMU, haptics)**
- 6. Power (boost, charger, fuel gauge, protection, LDOs)**
- 7. Battery**
- 8. USB-C + protection**
- 9. MIPI / camera ESD**
- 10. CM4 board-to-board connectors**
- 11. Other connectors + indicators**
- 12. Passives (summary)**

- **Generated:** 2026-06-30 | **Board:** ai_glasses_carrier_v1
- **Scope:** key devices with manufacturer part number + headline datasheet parameters, by subsystem (compute + Wi-Fi/BT, camera, audio + speaker, GNSS, sensors, power, battery, USB-C, protection, connectors).
- MPN + footprint are pulled live from the BOM ([scripts/carrier_bom.py](#)); the electrical/mechanical specs are curated from the manufacturer datasheets.

[!] **Headline figures only.** Verify exact numbers and the full orderable suffix (reel/package/grade) against the latest manufacturer datasheet before procurement. Wi-Fi/Bluetooth are integrated in the CM4 SKU (this is a SoM carrier, not chip-down).

1. Compute module + wireless (Wi-Fi / Bluetooth)

The SoC, RAM, eMMC and the Wi-Fi/BT radio all live ON the Radxa CM4 module (this is a SoM carrier, not chip-down). They reach the carrier only through the three DF40 B2B connectors. Bluetooth/Wi-Fi are therefore part of the CM4 SKU, not separate carrier components.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
-	Radxa CM4 (RM1 26-D4E32J0R35 W28)	Radxa	SoM, 3x 100-pin DF40 B2B	SoC Rockchip RK3576 (4x Cortex-A72 + 4x A53, NPU ~6 TOPS); RAM 4GB LPDDR4X; eMMC 32GB; supply 5V DCIN; GPIO bank 1.8V (GPIO_VREF)	B2B J31/J32/J1
-	Wi-Fi (on CM4 module)	on-module	in CM4 SKU	802.11ax (Wi-Fi 6), 2.4 + 5 GHz; antenna on-module / per SKU	SDIO/PCIe in CM4
-	Bluetooth (on CM4 module)	on-module	in CM4 SKU	Bluetooth 5.x (BLE + classic); shares CM4 wireless front-end; antenna per SKU	UART/PCM in CM4

2. Camera

Sony IMX415 (Radxa Camera 4K) on CM4 CSI3, 4-lane MIPI, via a 31-pin 0.3mm FFC.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
-	IMX415-AAQR-C (Radxa Camera 4K)	Sony	module (off-board, via J2 FFC)	1/2.8in 8.29 MP (3864x2192) CMOS; 4-lane MIPI CSI-2; up to 90 fps; rails IO 1.8V / analog 2.9V / core 1.1V (on module); I2C control	CSI3 4-lane, I2C0
J2	FH35C-31S-0.3S HW(50)	Hirose	`AI_Glasses:FH35 C-31S-0.3SHW_5 0`	31-pos 0.3mm-pitch FFC/FPC connector, bottom-contact; camera flex	CSI3 FFC

3. Audio (amplifier, microphones, speaker)

Class-D speaker path + dual PDM MEMS mics. Speaker is off-board via J3.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
U4	MAX98357AETE+ T	Analog Devices (Maxim)	`Package_DFN_Q FN:QFN-16-1EP_ 3x3mm_P0.5mm_ EP1.45x1.45mm`	Mono class-D amp; I2S/SAI (Left /Right/(L+R)/2); ~3.2W into 4ohm @5V (~92% eff); no MCLK needed; gain 3/6/9/12/15 dB; supply 2.5-5.5V	SAI1 (BCLK/LRCK/DIN)
MK1	SPH0641LU4H-1	Knowles	`Sensor_Audio:Kn owles_LGA-5_3.5 x2.65mm`	PDM digital MEMS mic, bottom port; SNR 64.3 dB(A); AOP 120 dB SPL; sens -26 dBFS; supply 1.6-3.6V; ultrasonic-capable	PDM1 (L, SEL=GND)
MK2	SPH0641LU4H-1	Knowles	`Sensor_Audio:Kn owles_LGA-5_3.5 x2.65mm`	Same as MK1; right channel	PDM1 (R, SEL=VDD)
-	OWS-091630W50 A-8 (speaker)	OWS / generic	9x16x3.0 mm (off-board via J3)	8 ohm; rated ~1 W (peak higher); micro rectangular speaker; SPL/Fo per datasheet	amp OUT+/OUT- (post FB1/FB2)
FB1	Ferrite 600ohm@ 100MHz	Murata/TDK class	`Inductor_SMD:L_ 0603_1608Metric`	Class-D EMI bead on OUT+; with C23 1nF	speaker EMI

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
FB2	Ferrite 600ohm@100MHz	Murata/TDK class	`Inductor_SMD:L_0603_1608Metric`	Class-D EMI bead on OUT-; with C24 1nF	speaker EMI

4. GNSS (positioning)

u-blox MAX-M10S over UART; external active antenna via U.FL.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
U14	MAX-M10S-00B-01	u-blox	`RF_GPS:ublox_MAX`	GNSS receiver GPS/GLONASS/Galileo/BeiDou/QZSS; tracking sens -167 dBm; ~25 mA continuous; VCC 2.7-3.6V, VIO 1.8V; UART/I2C	UART7, U.FL (J8)
J8	U.FL-R-SMT-1	Hirose	`Connector_Coaxial:U.FL_Hirose_U.FL-R-SMT-1_Vertical`	50 ohm U.FL/IPEX coax RF connector; external active GNSS antenna	RF 50 ohm
D4	low-cap RF ESD (<=0.3pF)	TBD (Nexperia/Littelfuse class)	`Diode_SMD:D_SOD-882`	RF ESD clamp on antenna feed; capacitance <=0.3 pF; Vrrwm >=5V	antenna ESD

5. Sensors (IMU, haptics)

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
U10	ICM-42688-P	TDK InvenSense	`Package_LGA:LGA-14_3x2.5mm_P0.5mm_LayoutB_order3x4y`	6-axis IMU; accel +/-2/4/8/16 g; gyro up to +/-2000 dps; gyro noise ~2.8 mdps/rHz; I2C/SPI; supply 1.71-3.6V	I2C8, INT1
U11	DRV2605L	Texas Instruments	`Package_SO:MSOP-10_3x3mm_P0.5mm`	Haptic driver for LRA/ERM; I2C; 123-effect waveform ROM (TI/Immersion); supply 2-5.2V	I2C + PWM (vib motor J6)

6. Power (boost, charger, fuel gauge, protection, LDOs)

1S Li-Po -> 5V system via boost; per-rail LDOs; charge + gauge + cell protection.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
U6	TPS61022	Texas Instruments	`Package_DFN_Q FN:Texas_RWU0 007A_VQFN-7_2x 2mm_P0.5mm`	Sync boost; Vin 0.5-5.5V; Vout up to 5.5V; 8A switch; up to 96% eff; ~1 MHz	VBAT -> +5V_SYS
L1	Boost inductor 4.7uH	TBD (Cyntec/Wurth class)	`AI_Glasses:POW ER_INDUCTOR_ 7X7_VERIFY`	4.7 uH; saturation >=10 A; low DCR; shielded; size per measured peak current	boost SW node
U7	BQ25180	Texas Instruments	`Package_BGA:T exas_DSBGA-12_ 1.36x1.86mm_Lay out3x4_P0.5mm`	1S Li-ion charger w/ power path; up to 1A charge; input 3.5-5.5V; I2C; ship mode	I2C6, USB-C 5V in
U8	BQ29700	Texas Instruments	`Package_TO_SO T_SMD:SOT-23-5 <td>Single-cell protection: OVP/U VP/OCP/SCP; ultra-low IQ</td> <td>battery protection</td>	Single-cell protection: OVP/U VP/OCP/SCP; ultra-low IQ	battery protection
U9	MAX17048	Analog Devices (Maxim)	`Package_DFN_Q FN:DFN-8-1EP_2 x3mm_P0.5mm_E P0.61x2.2mm`	ModelGauge fuel gauge (SOC); I2C; ~3 uA hibernate; low-batt alert	I2C6
U3	LP5907MFX-3.3	Texas Instruments	`Package_TO_SO T_SMD:SOT-23-5 <td>250 mA ultra-low-noise LDO; 6.5 uVrms; PSRR ~82 dB; -> camera 3.3V</td> <td>+CAM_3V3</td>	250 mA ultra-low-noise LDO; 6.5 uVrms; PSRR ~82 dB; -> camera 3.3V	+CAM_3V3
U12	TLV75733	Texas Instruments	`Package_TO_SO T_SMD:SOT-23-5 <td>1A LDO; low IQ; -> 3.3V system rail</td> <td>+3V3</td>	1A LDO; low IQ; -> 3.3V system rail	+3V3
U13	TLV75718	Texas Instruments	`Package_TO_SO T_SMD:SOT-23-5 <td>1A LDO; low IQ; -> 1.8V VCCIO rail</td> <td>+1V8</td>	1A LDO; low IQ; -> 1.8V VCCIO rail	+1V8
U15	TPS25940	Texas Instruments	`Package_DFN_Q FN:Texas_RGP00 20H_VQFN-20-1E P_4x4mm_P0.5m m_EP2.4x2.4mm`	eFuse: OVP/OCP/ reverse-block/inru sh; Vin 2.7-18V; optional, DNP by default	VBUS in

7. Battery

1S Li-Po. V1 bench cell locked; glasses build uses thin temple cells.

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
J5	S3B-PH-SM4-TB	JST	`Connector_JST:JST_PH_S3B-PH-SM4-TB_1x03-1MP_P2.00mm_Horizontal`	Battery connector BAT+/BAT-/NTC; 2.0mm pitch	battery
-	V1 bench cell (103450)	generic 1S Li-Po	10x34x50 mm	1S 3.7V nominal; ~2000 mAh; with PCM; NTC recommended	via J5
-	Glasses pack (target)	custom 1S	~35x18x4 mm cells	2-3 cells, ~250 mAh each (~500-750 mAh); >3-4h needs AI duty-cycling	via J5

8. USB-C + protection

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
J4	USB4085-GF-A	GCT	`Connector_USB:USB_C_Receptacle_GCT_USB4085`	USB-C receptacle; USB2 + 5V sink; 16-pos	USB2, 5V in
U5	TPD2E009	Texas Instruments	`Package_TO_SOT_SMD:SOT-553`	2-ch ESD clamp for USB2 D+/D-; low cap ~1.5 pF; V _{rw} >=5V	USB D+/D-
D3	SMAJ5.0A	Littelfuse/Bourns class	`Diode_SMD:D_SMA`	Uni TVS for VBUS; 5V standoff; 400W peak pulse	VBUS surge/ESD
R1	5.1k 1%	Yageo class	`Resistor_SMD:R_0603_1608Metric`	USB-C CC1 sink pull-down (locked 1%)	CC1
R2	5.1k 1%	Yageo class	`Resistor_SMD:R_0603_1608Metric`	USB-C CC2 sink pull-down (locked 1%)	CC2

9. MIPI / camera ESD

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
U2	TPD4E05U06	Texas Instruments	`Package_SON:U SON-10_2.5x1.0 mm_P0.5mm`	4-ch ultra-low-cap (0.4 pF) ESD array for MIPI CSI lanes; Vrrm >=5V	CSI3 lanes
U16	TPD4E05U06	Texas Instruments	`Package_SON:U SON-10_2.5x1.0 mm_P0.5mm`	Second 4-ch low-cap ESD array (remaining CSI lanes)	CSI3 lanes

10. CM4 board-to-board connectors

Carrier-side receptacles; mate the CM4's DF40C-100DP plugs; 1.5mm board gap. Footprint geometry is the KiCad library part; the locked 3-connector PLACEMENT still needs the mechanical fit-check (official CM4 STEP + 2-layer coupon).

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
J31	DF40C-100DS-0.4V(51)	Hirose	`Connector_Hirose_DF40:Hirose_DF40C-100DS-0.4V_2x50_P0.4mm`	100-pin, 0.4mm pitch B2B receptacle; CM4 J3A (low-speed)	CM4 J3A
J32	DF40C-100DS-0.4V(51)	Hirose	`Connector_Hirose_DF40:Hirose_DF40C-100DS-0.4V_2x50_P0.4mm`	100-pin, 0.4mm pitch B2B receptacle; CM4 J3B (high-speed: MIPI/USB)	CM4 J3B
J1	DF40C-100DS-0.4V(51)	Hirose	`Connector_Hirose_DF40:Hirose_DF40C-100DS-0.4V_2x50_P0.4mm`	100-pin, 0.4mm pitch B2B receptacle; CM4 J1 (I/O: SAI, MCLK)	CM4 J1

11. Other connectors + indicators

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
J3	SM02B-SRSS-TB	JST	`Connector_JST:JST_SH_BM02B-SRSS-TB_1x02-1MP_P1.00mm_Vertical`	Speaker connector (post-EMI); 1.0mm pitch	speaker

Ref	Part / MPN	Mfr	Package	Key parameters	Interface / Rail
J6	SM02B-SRSS-TB	JST	`Connector_JST:JST_SH_BM02B-SRSS-TB_1x02-1MP_P1.00mm_Vertical`	Vibration motor connector; 1.0mm pitch	haptic
J7	Header 1x4 2.54mm	generic	`Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical`	UART debug (TX/RX/GND/VREF)	UART0
D1	LED Green	generic	`LED_SMD:LED_0603_1608Metric`	Power LED; active-low GPIO sink; via R4 1k	status
D2	LED Blue	generic	`LED_SMD:LED_0603_1608Metric`	Status LED; active-low GPIO sink; via R5 1k	status

12. Passives (summary)

Full per-family passive audit (value / tolerance / dielectric / voltage / DC-bias verdicts) is in [AI_Glasses_Carrier_V1_Design_Review.pdf §7](#). Headline:

Family	Refs (examples)	Spec
I2C pull-ups	R7-R13	2.2k 1% 0603
USB-C CC	R1, R2	5.1k 1% 0603 (locked)
Decoupling	C2,C4,C6,C10,C13...	100nF X7R 50V 0603
Bulk MLCC	C1,C3,C5,C8,C15,C29	10uF X5R 25V 0805
VBAT bulk	C7	22uF X5R 25V 0805
Class-D EMI	C23, C24	1nF X7R 50V 0603
Current shunt	R3	mohm 2512 (select for 5V measure)

_Generated by [scripts/generate_parts_spec.py](#). MPN/footprint live from the BOM; headline specs curated from manufacturer datasheets - verify before procurement._